Pharmaceutical Analytical Chemistry II
PCC/PCD 105

Week 3 lecture

Oxidation Reduction Reactions

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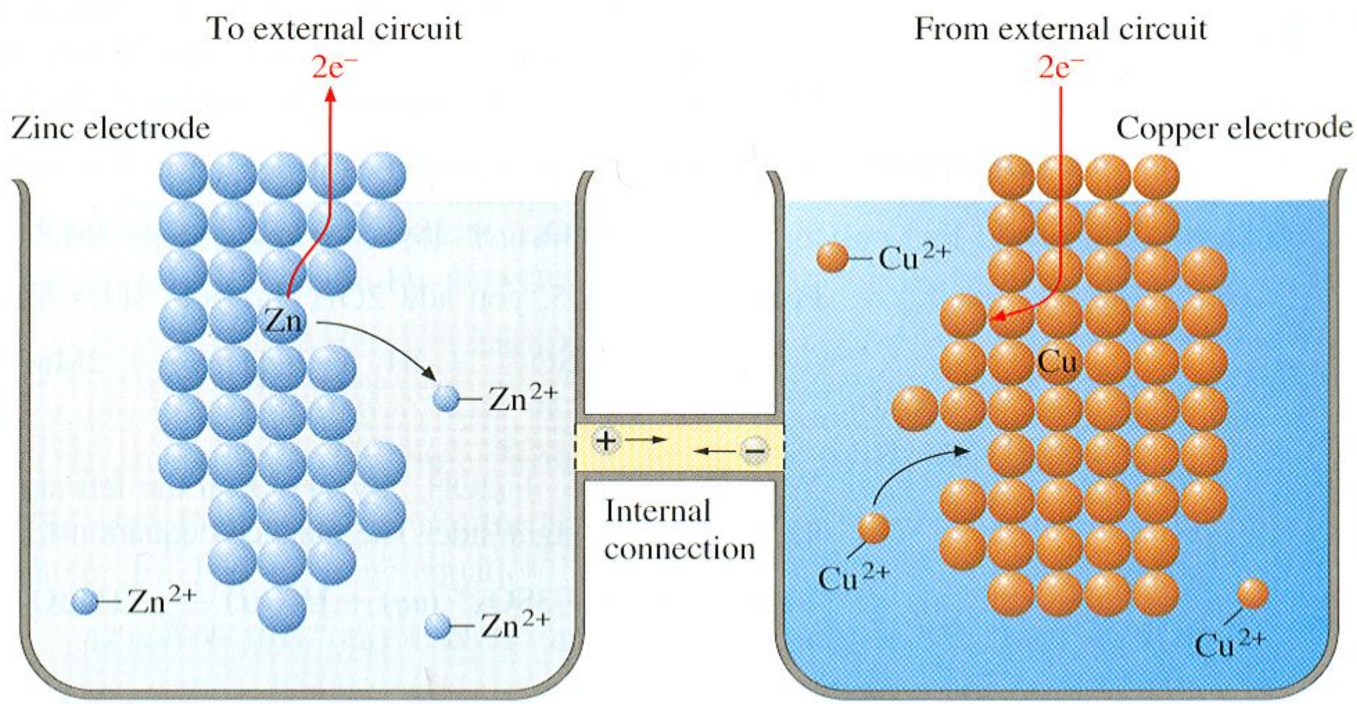
Oxidation Reduction Reactions



Electrode Potential

"Half-Cell Potential" "Single electrode potential"

The potential that develops at the phase boundary between Electrode / Electrolyte solution

It is **impossible** to determine *absolute* half-cell potentials, we can readily determine *relative* electrode potentials, *because* all voltage-measuring devices measure only *differences* in potential.

Cell Potential

The *difference* between two half-cell potentials

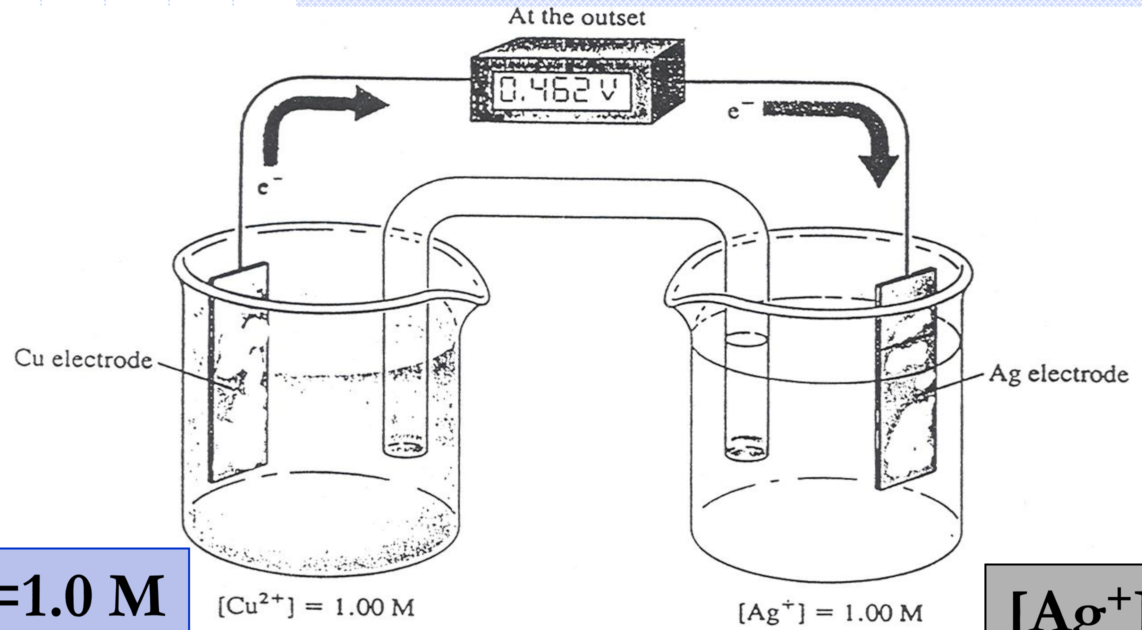
$$E_{cell} = E_{cathode} - E_{anode}$$

"+… galvanic"
"- … Electrolytic"

It is potential difference that develops between the Electrodes of the cell.

***Considered as a measure of the tendency for the reaction**

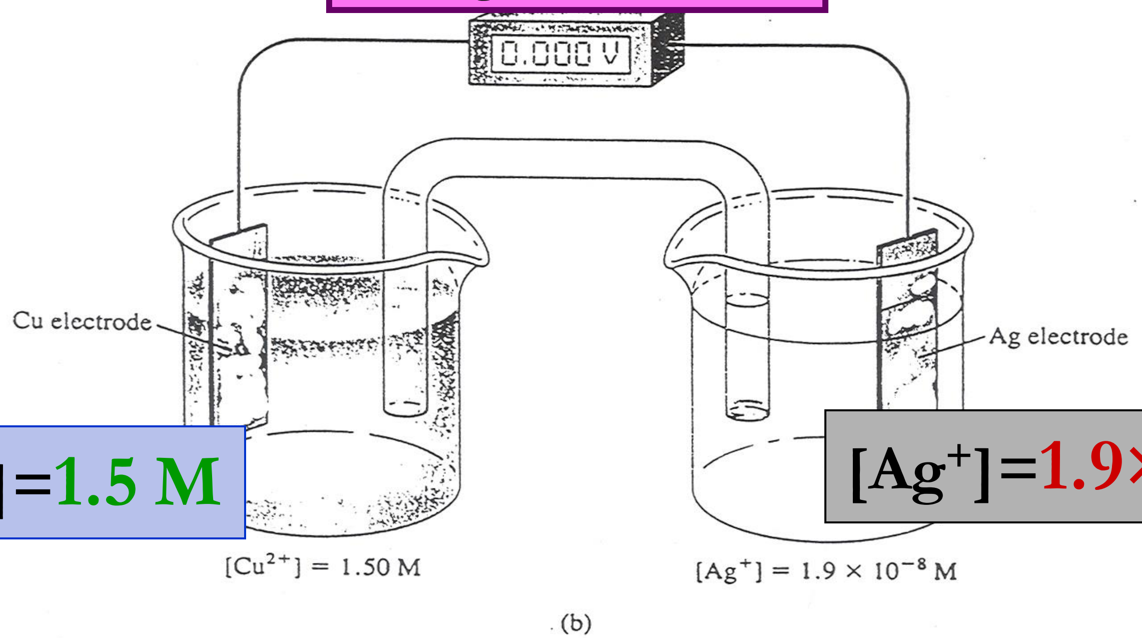
Figure3: Change in Cell Potential After Passage of Current



$[Cu^{2+}]$=1.0 M

$[Ag^{+}]$=1.0 M

AT EQUILIBRIUM



$[Cu^{2+}]$=1.5 M

$[Ag^{+}]$=1.9×10^{-8} M

- According to Cell Rx: $2\,Ag^{+} + Cu(s) \Leftrightarrow 2\,Ag(s) + Cu^{2+}$
- As shown in Figure-3(a), when the copper and silver ion activities in the cell are **1.00 M**, a potential of **0.462 V develops**, which shows that this reaction is far from equilibrium.

- As the reaction proceeds, this **potential becomes smaller** until **at equilibrium** the **meter reads** **0.00 V**. As shown in Figure-3(b), the **copper** ion **equilibrium concentration** is then just slightly less than **1.5 M** (increased than the starting concentration) and the **silver** ion concentration is **1.9×10^{-8} M "very low conc."**.

- At this point, no further net flow of electrons will occur.

- Although we **cannot** determine **absolute** potentials of electrodes, we **can** readily determine ***relative*** electrode potentials.

For example, if we replace the copper anode in Figure-1(a) with a cadmium electrode immersed in a cadmium sulfate solution, the voltmeter reads about 0.7 V greater than the original cell.

Define:

- **Liquid junction potential**
- **Electrode potential**
- **Relative electrode potential**
- **Cell potential**

Schematic representation

$Cu/CuSO_4$ Anode $\quad$ $Ag/AgNO_3$ Cathode

$$Cu \,|\, Cu^{2+}\ (1.0\ M) \;||\; Ag^{+}\ (1.0\ M) \,|\, Ag$$

$$Cu \,|\, CuSO_4\ (1.0\ M) \;||\; AgNO_3\ (1.0\ M) \,|\, Ag$$

Example:

$Zn/ZnSO_4$ Anode $\quad$ $Cu/Cu\ SO_4$ Cathode

$$Zn \,|\, Zn^{2+}\ (1.0\ M) \;||\; Cu^{2+}\ (1.0\ M) \,|\, Cu$$

$$Zn \,|\, ZnSO_4\ (1.0\ M) \;||\; CuSO_4\ (1.0\ M) \,|\, Cu$$

A single vertical line indicates a phase boundary or interface at which a potential develops "Electrode Potential"

$Zn \,|\, ZnSO_4$ Anode *potential: Potential* develops at a phase boundary between Zinc electrode and Zinc sulfate solution

A double vertical line represents the salt bridge

Anode is assigned as left-hand electrode

Cathode is assigned as right-hand electrode

( ) activity

TYPES of ELECTRODES "Half-Cell"

1. INERT Electrode:

It does not share in electrode reaction. It acts as a conductor of electrons ·

Example: Pt, gold,graphite,... it is used in the following electrodes:

- Gas electrode,
  E.g. Hydrogen electrode H_2 gas/H^+

  Pt, H_2 (p = ? atm) | H^+ (1.00 M) ||

- Metal ions in different oxidation states Fe^{3+}/ Fe^{2+}

  Pt | Fe^{3+} (?M) , Fe^{2+} (?M) ||

- Non-Metal ions in different oxidation states

  Pt | ClO_4^- (?M), Cl^- (?M) ||

2. METAL / METAL ION Electrode,

Ag/Ag^+ Ag | Ag^+ (x M) ||

Zn/Zn^{2+} Zn | Zn^{2+} (x M) ||

3. METAL / INSOLUBLE SALT OF METAL / ION IN COMMON WITH SALT electrode

- Silver/silver chloride electrode:

  $Ag/AgCl/Cl^-$ Ag | AgCl (sat´d), Cl^- (x M) ||

- Calomel electrode: $Hg/ Hg_2Cl_2/Cl^-$ Hg | Hg_2Cl_2 (sat´d), Cl^- (x M) ||

Standard electrode is the ectrodee at standard conditions:

1- Conc. of ions in aq. solutions = 1M

2- Pressure of gases = 1 atm. (760 mm Hg)

3- temp. = 25° C (298 K)

E°.. Standard electrode potential..

Temperature dependent

N.B. E.. electrode potential (not standard)

The Standard Hydrogen Reference Electrode

- has been used throughout the world for many years as a universal reference electrode.
- It is a **typical *gas electrode***.
- Iis ***reversible*** and acts either as **an anode** or as a **cathode**, depending upon the half-cell with which it is coupled.

$$2\,H^{+}\,(aq) \;+\; 2\,e^{-} \;\Leftrightarrow\; H_2(g)$$

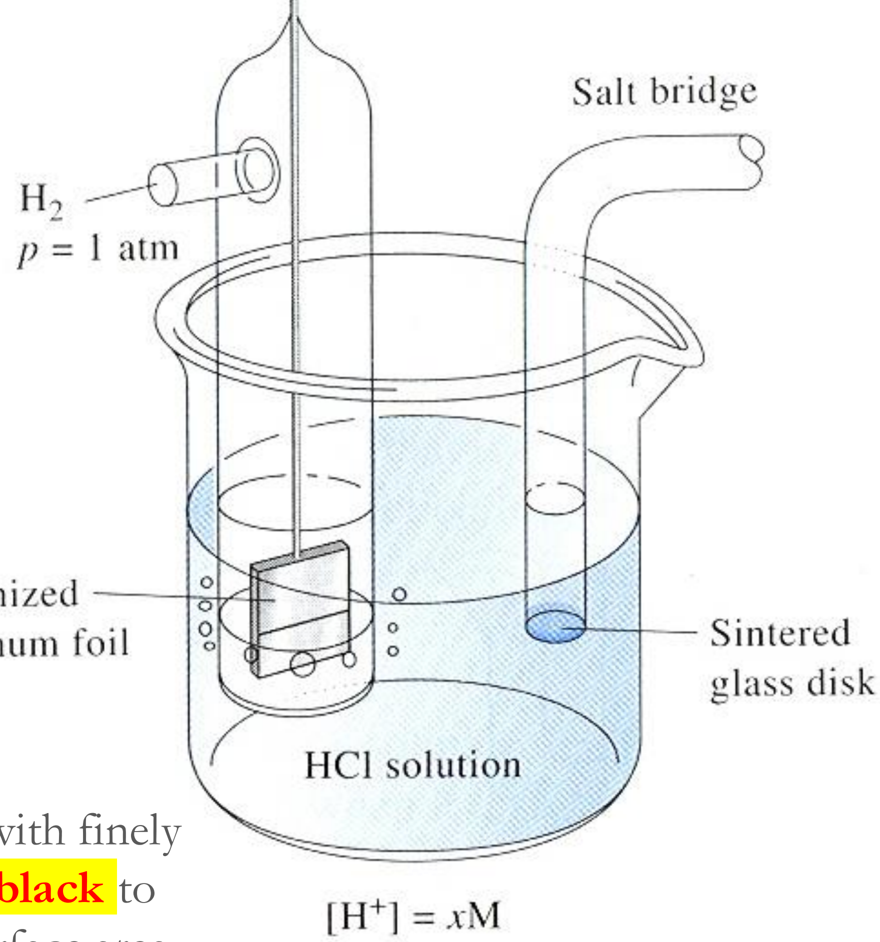
Hydrogen gas is oxidized to hydrogen ion when this electrode serves as the anode.

Hydrogen ion is reduced to molecular hydrogen when the hydrogen electrode acts as a cathode.

- *By convention, the potential of the standard hydrogen electrode is assigned a **value of zero** at all temperatures*.

Hydrogen Gas Electrode

$$2\,H^+\,(aq) + 2\,e^- \Leftrightarrow H_2(g)$$



"Metal conductor"

Platinized with finely divided **Pt black** to increase surface area

Definition of Electrode Potential

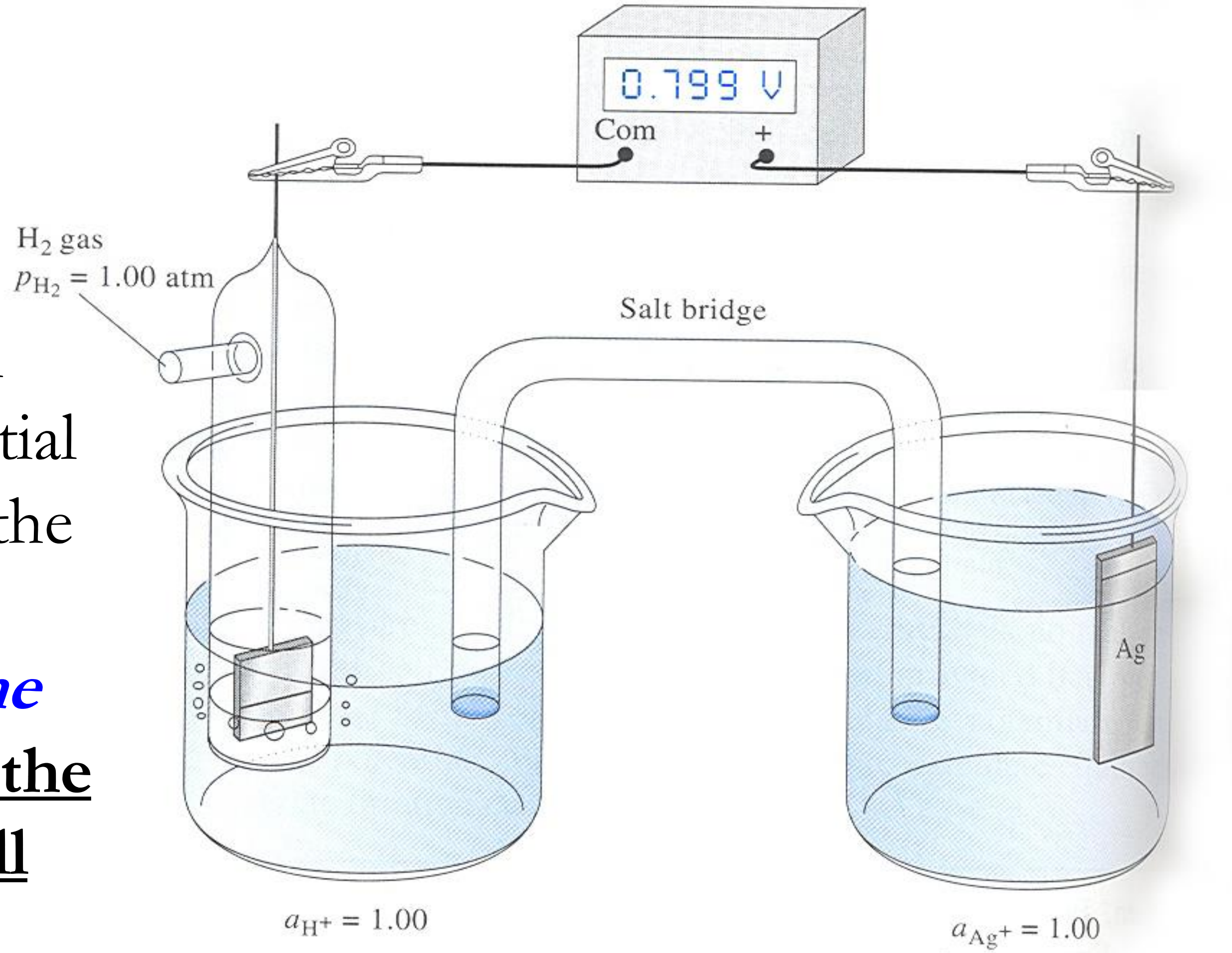
- An *electrode potential* is <u>defined as</u> the potential of a cell consisting of the electrode in question *acting as a cathode* and the **<u>standard hydrogen electrode</u>** *acting* ***<u>as an anode</u>*** ….. “**relative electrode potential**”.

- For

$$\mathrm{SHE} \,||\, Ag^{+}\,(0.02\ M)\,|\, Ag$$

$$E_{cell} = E_{Ag} - 0.000 = E_{Ag}$$

for $Ag^+ + e^- \Leftrightarrow Ag(s)$ $E_{Ag+} = +\ 0.799\ V$



This galvanic cell develops a potential of 0.799 V with the silver electrode functioning ***as the cathode***; that is, **the spontaneous cell reaction is**:

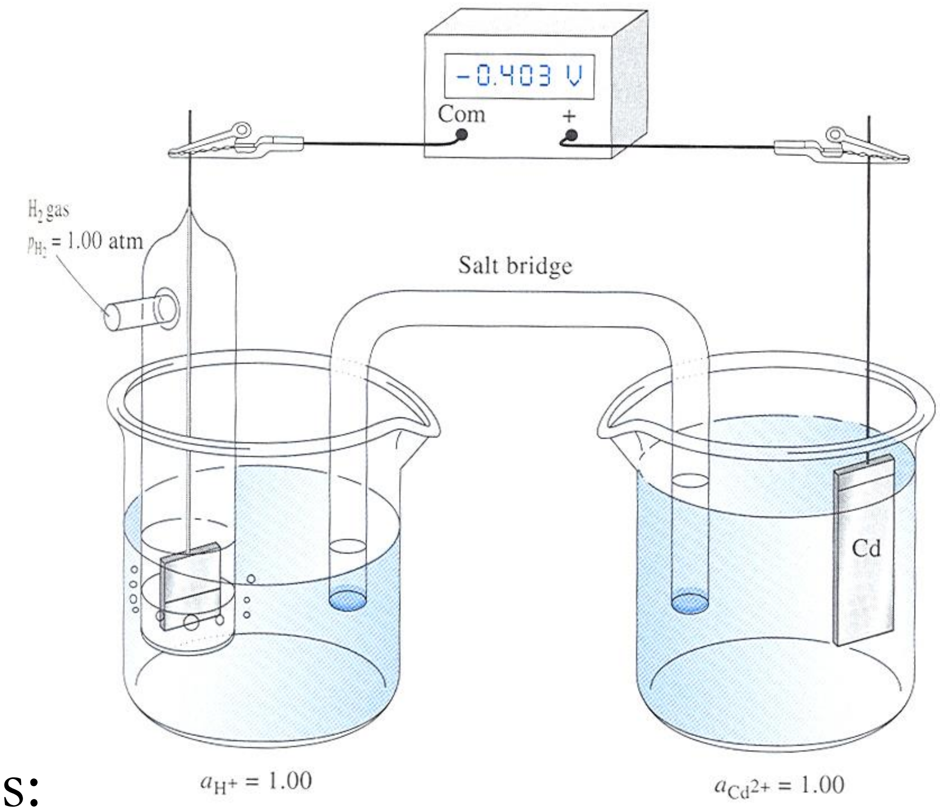
$$2\ Ag^+ + H_2(g) \Leftrightarrow 2\ Ag(s) + 2H^+$$

For

$$Cd^{2+} + 2e^- \Leftrightarrow Cd(s) \quad E_{Cd^{2+}} = -0.403\ V$$

The **cadmium electrode** bears a **negative charge** with respect to the standard hydrogen electrode.

Zinc electrode functioning ***as the anode***;



The **spontaneous** cell reaction is:

$$Cd(s) + 2H^+ \Leftrightarrow H_2(g) + Cd^{2+}$$

Half-Reaction	Electrode Potential, V
$Ag^+ + e^- \Leftrightarrow Ag(s)$	+ 0.799
$2\,H^+ + 2\,e^- \Leftrightarrow H_2(g)$	0.000
$Cd^{2+} + 2\,e^- \Leftrightarrow Cd(s)$	- 0.403
$Zn^{2+} + 2e^- \Leftrightarrow Zn(s)$	- 0.763

The magnitudes of electrode potentials indicate the relative strengths of the ionic species as **electron acceptors** (**oxidizing agents**); that is, in **decreasing** strength, undergoes reduction "i.e. as E increases, tendency for reduction increases, more powerful oxidizing agent"

$$Ag^+ > H^+ > Cd^{2+} > Zn^{2+}$$

TABLE 18-1

Standard Electrode Potentials*

Reaction	E^0 at 25°C, V
$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-$	+1.359
$O_2(g) + 4H^+ + 4e^- \rightleftharpoons 2H_2O$	+1.229
$Br_2(aq) + 2e^- \rightleftharpoons 2Br^-$	+1.087
$Br_2(l) + 2e^- \rightleftharpoons 2Br^-$	+1.065
$Ag^+ + e^- \rightleftharpoons Ag(s)$	+ 0.799
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+ 0.771
$I_3^- + 2e^- \rightleftharpoons 3I^-$	+ 0.536
$Cu^{2+} + 2e^- \rightleftharpoons Cu(s)$	+ 0.337
$UO_2^{2+} + 4H^+ + 2e^- \rightleftharpoons U^{4+} + 2H_2O$	+ 0.334
$Hg_2Cl_2(s) + 2e^- \rightleftharpoons 2Hg(l) + 2Cl^-$	+ 0.268
$AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-$	+ 0.222
$Ag(S_2O_3)_2^{3-} + e^- \rightleftharpoons Ag(s) + 2S_2O_3^{2-}$	+ 0.017
$\mathbf{2H^+ + 2e^- \rightleftharpoons H_2(g)}$	**0.000**
$AgI(s) + e^- \rightleftharpoons Ag(s) + I^-$	− 0.151
$PbSO_4 + 2e^- \rightleftharpoons Pb(s) + SO_4^{2-}$	− 0.350
$Cd^{2+} + 2e^- \rightleftharpoons Cd(s)$	− 0.403
$Zn^{2+} + 2e^- \rightleftharpoons Zn(s)$	− 0.763

*See Appendix 5 for a more extensive list.

The sign Convention for Electrode Potentials

- The **<u>sign of an electrode</u>** potential is determined by the *<u>sign of the half-cell in question when it is coupled to a standard hydrogen electrode</u>*.
- When the half-cell of interest behaves **spontaneously as a cathode**, it is the positive electrode of the galvanic cell. Thus, its electrode potential is **<u>positive</u>**.
- When the half-cell of interest behaves **as an anode**, the electrode is **<u>negative</u>** Thus, its electrode potential is **<u>negative</u>**.
- It is important to emphasize that <u>the electrode potential refers to a **half-cell process** written ***as a reduction***</u>.
- For the **zinc** and **cadmium** electrodes, the spontaneous reactions are oxidations. (spontaneous anodes)

- *It is evident, then, that the sign of an electrode potential indicates **whether the reduction is spontaneous** with respect to the **standard hydrogen electrode**.*
- The **positive sign** associated with the electrode potential for silver indicates that the process:

$$2\,Ag^+ + H_2(g) \Leftrightarrow 2\,Ag(s) + 2\,H^+$$

favors the products under ordinary conditions.

Ag electrode is spontaneous cathode

$$Zn^{2+} + H_2(g) \Leftrightarrow Zn(s) + 2\,H^+$$

- Similarly, the **negative sign** of the electrode potential for zinc means that the analogous reaction, is **not spontaneous**. That is, the reaction tends to go in the other direction:

$$Zn(s) + 2\,H^+ \Leftrightarrow Zn^{2+} + H_2(g)$$

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$Cd^{2+} + 2e^- \rightleftharpoons Cd(s)$	− 0.403
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*See Appendix 5 for a more extensive list.

- Consider the half-reaction:

$$aA + bB + ne^- \Leftrightarrow cC + dD$$

$$E = E^0 - \frac{0.0592}{n} \log \frac{[C]^c [D]^d \ldots}{[A]^a [B]^b \ldots} = \frac{[\text{Products}]}{[\text{Reactants}]}$$

- The above Equation is known as the ***Nernst** equation*

(1) $Zn^{2+} + 2\,e^- \Leftrightarrow Zn(s)$ $\qquad E = E^0 - \frac{0.0592}{2} \log \frac{1}{[Zn^{2+}]}$

(2) $Fe^{3+} + e^- \Leftrightarrow Fe^{2+}$ $\qquad E = E^0 - \frac{0.0592}{1} \log \frac{[Fe^{2+}]}{[Fe^{3+}]}$

(3) $2\,H^+ + \mathbf{2\,e^-} \Leftrightarrow H_2(g)$ $\qquad E = \boldsymbol{E^0} - \dfrac{0.0592}{\mathbf{2}} \log \dfrac{pH_2}{[H^+]^2}$

4) $MnO_4^- + \mathbf{5\,e^-} + 8\,H^+ \Leftrightarrow Mn^{2+} + 4\,H_2O$

$$E = \boldsymbol{E^0} - \frac{0.0592}{\mathbf{5}} \log \frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8}$$

Cell Potential Calculations

$$E_{cell} = E_{cathode} - E_{anode}$$

Example-1:

Calculate the thermodynamic potential of the following cell:

$$\mathbf{Cu \mid Cu^{2+}(0.100\ M) \parallel Ag^{+}(0.200\ M) \mid Ag}$$

The two half-reactions and standard potentials are

$Ag^+ + e^- \Leftrightarrow Ag(s)$ $\qquad$ $E^0 = \mathbf{0.799\ V}$

$Cu^{2+} + 2e^- \Leftrightarrow Cu(s)$ $\qquad$ $E^0 = \mathbf{0.337\ V}$

The electrode potentials are,

$$E_{Ag^+} = \mathbf{0.799} - 0.0592 \log \frac{1}{0.200} = 0.758\ V$$

$$E_{Cu^{2+}} = \mathbf{0.337} - \frac{0.0592}{2} \log \frac{1}{0.100} = 0.307\ V$$

$$E_{cell} = E_{Ag^+} - E_{Cu^{2+}} = 0.758 - 0.307 = \mathbf{+\ 0.451}$$

Galvanic Cell-Spontaneous Rx

Example -2:

Calculate the thermodynamic potential for the cell

$$\mathbf{Ag\,|\,Ag^{+}(0.200\ M)\,||\,Cu^{2+}(0.100\ M)\,|\,Cu}$$

The electrode potentials for the two half-reactions are identical to the electrode potentials calculated in Example-4. That is,

$$E_{Ag^{+}} = 0.758\ V \qquad \text{and} \qquad E_{cu^{2+}} = 0.307\ V$$

In **contrast to** the previous example, however, the **silver electrode** is the **anode** and the **copper electrode** the **cathode**.

Substituting these electrode potentials into Equation-10 gives:

$$E_{cell} = E_{cu^{2+}} - E_{Ag^{+}} = 0.307 - 0.758 = \mathbf{-\ 0.451\ V}$$

Electrolytic Cell- non Spontaneous Rx

Example-3:

Calculate the potential of a silver electrode immersed in a 0.0500 M solution of NaCl and saturated AgCl(s) using,

(a) $E^0_{Ag^+} = 0.799$ V and **(b) $E^0_{AgCl} = 0.222$ V.**

(a) $Ag^+ + e^- \Leftrightarrow Ag(s) \qquad E^0 = 0.799$ V

The Ag^+ concentration of this solution is given by

$[Ag^+] = K_{sp}/[Cl^-] = 1.82 \times 10^{-10}/0.0500 = 3.64 \times 10^{-9}$ M

Substituting into the Nernst expression gives

$$E = 0.799 - 0.0592 \log \frac{1}{3.64 \times 10} = 0.299 \text{ V}$$

(b) Here we may write, $AgCl + e^- \rightarrow Ag + Cl^-$

$E = 0.222 - 0.0592 \log [Cl^-] = 0.222 - 0.0592 \log 0.0500 = 0.299$ V